AMRITA VISHWA VIDYAPEETHAM
DEEMED TO BE UNIVERSITY UNDER SECTION 3 OF UGC ACT, 1956
COIMBATORE CAMPUS

INSTITUTION'S INNOVATION COUNCIL
(Ministry of Education Initiative)

GENESIS

Genesis Ideathon: Final Presentation

Team ID: *GEN26-A0H-I115*

Team Name: *TEAM PR*

Domain: *Healthcare, Biotechnology and Human Well-being*

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Date: 11th October 2026

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Problem Statement

Patients often cannot access their medical history when: they cannot communicate (unconscious, in pain, language barrier), internet or power is down, they lack identity documents, or they are new to a facility. ABDM/ABHA provide large-scale linkage but assume connectivity, patient communication, or pre-existing linked records. In emergency settings this leads to medication errors, missed conditions (e.g., pregnancy, allergies), duplicated tests, and breaks in continuity of care.

Problem Understanding

- Who experiences this problem?
  1]Villagers who go to different clinics over time
  2]Old people who forget their medicines and can't explain their history
  3]Diabetic/hypertensive patients whose records are scattered across hospitals
  4]Accident victims who come in unconscious, with no ID and no family
  5]Rural folks who don't understand English or smartphones
  6]Doctors and nurses in small clinics who have to treat them

- What causes the problem?
  1]Paper records get lost or forgotten at home
  2]Every hospital has its own separate system that doesn't talk to others
  3]Internet goes down in villages exactly when it's needed
  4]People don't know how to use ABHA or digital apps
  5]Most apps are in English, not their local language

- How does it affect users or stakeholders?
  1] Patients get wrong or delayed treatment because their history is missing
  2]Same questions asked again and again at every clinic
  3]Rural hospitals freeze when internet drops during emergencies

Existing Gaps & Challenges

- limitation 1
  => Bureaucratic firewalls block medical data access
  Even with digital consent systems, hospitals require explicit written authorization to share records. If a patient is unconscious, family isn't present, or paperwork is delayed, doctors cannot legally access records—creating dangerous information blackouts during emergencies.

- limitation 2
  =>Rural clinics have no offline "paper backup" for digital data.
  ABHA stores records centrally online, but rural clinics rarely update records offline due to storage limits. If internet fails, clinics lose access to any digital records

- Main challenge that remains unsolved
  Most rural clinics lack IT staff to maintain complex systems.
  Expanding ABHA-linked access necessitates 100 percent uptime of clinic servers, which 70 percent of rural PHCs lack. Critical flaw: Enabling even one specialist to review records triggers multiple authorization steps that break during connectivity loss, nullifying the solution's emergency utility.

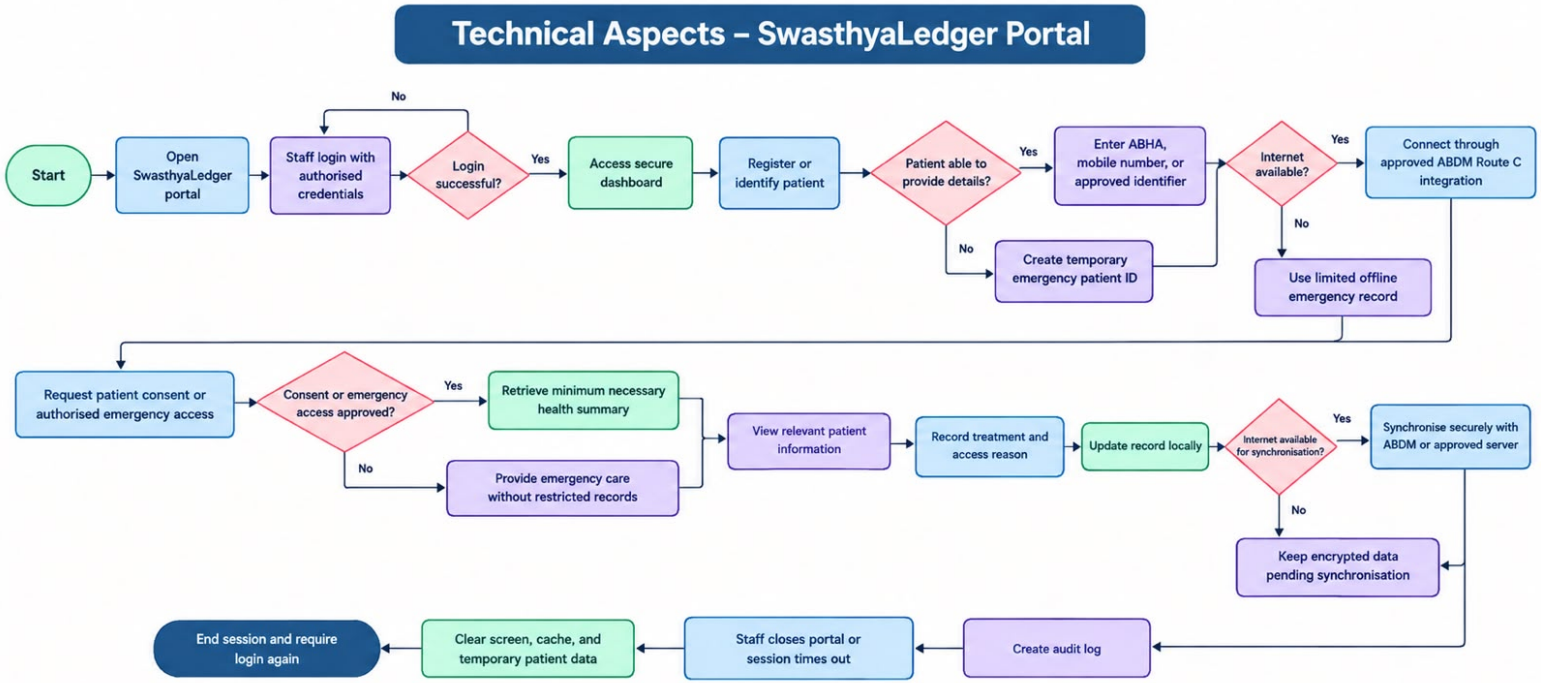
Our Proposed Solution

- Core idea or product
  A localized 'offline-first' health identity and emergency record access tool. It does not claim to unify India's whole medical history; instead, it acts as a practical, manageable bridge for rural clinics to confirm identity and retrieve relevant, pre-authorized health information safely during emergencies.

- How the solution works at a high level
  Scan: A nurse uses a fingerprint or face scan to check if the patient is in the local system.
  See: If there's a match, the doctor instantly gets key info—allergies, medications, past conditions.
  Fallback: If no match or no internet, the system creates a temporary record so treatment doesn't stop.

- Main value delivered to users
  Doctors: Treat patients faster and safer with the right info at hand.
  Patients: Get the care they need, even if they're unconscious or far from home.
  Clinics: Keep working smoothly, even when the internet cuts out.

Technical Approach



Tech Stack

Frontend: React + TypeScript (PWA) + Vite + Tailwind CSS.

Backend: Supabase (Postgres) or Node.js + PostgreSQL.

Database: IndexedDB for quick local storage on the tablet + SQLite for deeper records.

AI/ML, APIs, or Hardware: Iris and fingerprint scanning via the tablet's camera. voice commands in regional language

Innovation & Uniqueness

- What is new or different?
  SwasthyaLedger shifts the focus from patient self-identification to patient recognition in urgent care. It is designed for cases where the patient cannot communicate and the system still needs to help healthcare workers quickly identify them and access the right information.

- Why is this approach suited to the problem?
  It fits real healthcare conditions where time is limited, connectivity may fail, and staff need something simple and dependable. The offline-first and biometric-based approach reduces delays and makes the system usable when it matters most.

- What is the key differentiator?
  Its key differentiator is that it bridges the gap between existing health records and emergency care at the point of need. It is not trying to replace current systems; it makes them more usable in high-pressure situations.

Feasibility & Scalability

- **Feasibility:** Building this requires a small, focused team of developers—one for the mobile interface and one for the secure backend—along with low-cost tablets equipped with decent cameras. The path forward is practical: first, we create the offline database and iris-recognition tool, then we test it in a single clinic to ensure it feels natural for nurses to use, and finally, we connect it to the national health network (ABHA).

- **Scalability:** The system is designed to grow effortlessly because it lives on the devices themselves. We can start in one small clinic and expand to a whole state without needing massive, expensive servers at every location. As more clinics join, the "safety net" for patients gets stronger, eventually linking with private hospitals and pharmacies to create a truly seamless medical history that follows the patient everywhere.

- **Risks:** The biggest challenges are keeping patient data private and ensuring the tech stays simple enough for busy staff. We handle this by using high-level encryption—so even if a tablet is stolen, the data is useless—and by using voice commands so doctors don't have to waste time typing.

Use Cases

- Use case 1:
  The Unconscious Emergency Patient Our solution aims to transform how clinics handle trauma cases where every second counts. When an unidentified accident victim arrives unconscious, our system enables the nurse to scan an iris and instantly retrieve critical health alerts—like severe allergies or blood type—without waiting for family, papers, or internet. This gives doctors the certainty they need to act fast and safely.

- Use case 2:
  Rural Clinic with Zero Connectivity Our solution aims to keep rural healthcare flowing even when the world goes offline. In remote villages where internet fails for days, our system lets clinic staff record visits and update patient records locally. Once connectivity returns—perhaps when the nurse visits town—the data syncs seamlessly with national records. No missed updates, no lost information, just continuity of care.

Use Cases

- Use case 3:
  Elderly Patient with Cognitive Decline Our solution aims to protect vulnerable patients who cannot advocate for themselves. When an elderly person with memory loss arrives confused and unable to recall their medications, our system uses a simple fingerprint scan to pull up their complete profile and read it aloud. Doctors get the truth they need, and the patient is treated with both safety and dignity.

Impacts and Benefits

- Benefit to the target users:
  Our solution empowers healthcare workers to make faster, safer decisions without hunting for records. For unconscious, elderly, or non-communicative patients, their medical history is instantly available, preventing life-threatening errors like allergic reactions or drug interactions. Rural patients with chronic conditions gain continuity of care across clinics without losing their medical story.

- Social, economic, or environmental impact: Our solution ensures that a farmer in a village or an elderly person without a smartphone receives the same quality of care as someone in a city hospital. By preventing dangerous medical mistakes and unnecessary repeat tests, we ease the financial strain on families already struggling to afford healthcare.

- Expected measurable outcome:
  Within a 6-month pilot across 5 rural clinics, we aim to achieve: (1) 80 percent reduction in time spent retrieving patient history, (2) Zero medication-related errors in emergency cases, (3) 90 percent successful offline patient identification in low-connectivity areas

Sustainable Development Goals (SDGs)

SDG Number	Target Number	Goal Name	Connection
SDG [3]	Target [3.8]	[Universal health coverage]	[Ensures remote patients access quality healthcare without extra costs.]
SDG [10]	Target [10.2]	[Social and economic inclusion]	[Allows elderly and low-literacy patients easy access to their records, ensuring everyone can receive care.]

Individual Member Contribution

Member	Contribution or responsibility
Hanumithan A L	Analyzed how the solution aligns with existing health systems like ABHA/ABDM.
Nivedh T N	Researched technological solutions for patient record retrieval.
Shruthi Ranjane B S	Explored offline solutions for areas with limited connectivity.
Sunil Krishna G	Researched the problem of rural healthcare access and identified gaps.

Yukti ID

IR2026-1051085